Casual Document for Quantitative Mixing Time Results for Single Site Gibbs in Stochastic Block Model

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Contents

1 The Stochastic Block Model and the Single Site Gibbs Method **2**

Chapter 1

The Stochastic Block Model and the Single Site Gibbs Method

1.1 Basic Setup

The procedure of analyzing graphical data using Bayesian stochastic block model with single site Gibbs procedure seems to have been first mentioned in Kemp et al. (2004) and slightly later in Xu et al. (2006). In this section, we describe the simplest setup of both the model and the single site Gibbs method by following a tutorial article Schmidt and Mørup (2013).

For the moment, let us consider the simplest case where the graphical data received by the statistician is just one simple graph of $N \geqslant 3$ vertices labeled as $1, 2, \ldots, N$, represented by its adjacency matrix $X \in \{0,1\}^{N \times N}$ where $X_{i,j} = 1$ if and only if there is an edge between vertex i and vertex j. In particular, we have $X_{i,i} = 0$ for all i and $X_{i,j} = X_{j,i}$ for all i, j. If we want to, we can consider later the case which our data contains multiple graphs of the same size generated independently identically distributed from a distribution in the same model, but we shall not do so in this section.

We assume that X is generated from a distribution in the (standard Bayesian) stochastic block model (SBM) with unknown number of blocks. To describe the model, let us write Π for the collection of all partitions of the set $\{1, 2, \ldots, N\}$ and represent any element $z \in \Pi$ using the *restricted growth string* notation, meaning that we see z as a vector $z = (z_1, z_2, \ldots, z_N)$ where z_i denotes the block to which vertex i belongs, such that the classes are ordered as they first appear: $z_1 = 1$ and, for every $i \in \{2, 3, \ldots, N\}$, we have

$$z_i \in \left\{1, 2, \ldots, 1 + \max_{j<i} z_j\right\}.$$

For example, if $N = 5$ and the partition is $z = \{\{1, 3\}, \{2\}, \{4, 5\}\}$, then we write $z = (1, 2, 1, 3, 3)$. As derived in Schmidt and Mørup (2013), (the standard Bayesian) SBM is a parameteric model for the matrix X whose only parameter is z, ranged over Π, with density function is given by

$$p(X \mid z) = \prod_{1 \leqslant k \leqslant l \leqslant K} \frac{B(m_{k,l} + a, \bar{m}_{k,l} + b)}{B(a, b)}, \tag{1.1}$$

where $a > 0$ and $b > 0$ are hyperparameters fixed by the statistician, $B(\cdot, \cdot)$ is the beta function, K is the number of non-empty blocks as specified by z, $m_{k,l}$ is the number of edges between block k and block l, and $\bar{m}_{k,l}$ is the number of non-edges between block k and block l.

(Actually, just as in Schmidt and Mørup (2013), we have abused the notation a bit in (1.1). Given that X is symmetric with diagonal entries equal to 0, we are only interested in the vector $(X_{i,j})_{1 \leqslant i < j \leqslant N}$ and we have written $p(X \mid z)$ in place of $p((X_{i,j})_{1 \leqslant i < j \leqslant N} \mid z)$.)

In order to do Bayesian statistics, we have to specify a prior distribution for z. As pointed out in Schmidt and Mørup (2013), we will use the Chinese restaurant process (CRP) prior for z. With a hyperparameter $A > 0$

fixed by the statistician, CRP(A) is a probability measure on Π defined as follows: $z_1 \equiv 1$ and, for every $i \in \{2, 3, \ldots, N\}$, given $z_1, z_2, \ldots, z_{i-1}$ who have already used a total of $K^{(i-1)}$ blocks, z_i will join an existing block $k \in \{1, 2, \ldots, K^{(i-1)}\}$ with probability

$$\frac{n_k^{(i-1)}}{A + i - 1},$$

where $n_k^{(i-1)}$ is the number of vertices among $1, 2, \ldots, i-1$ who have been assigned to block k, and will create a new block $K^{(i-1)} + 1$ with probability

$$\frac{A}{A + i - 1}.$$

CRP(A) does have a clean joint density (with respect to the counting measure on Π)

$$p(z) = \frac{A^K \Gamma(A)}{\Gamma(A + N)} \prod_{k=1}^{K} \Gamma(n_k), \tag{1.2}$$

where K is the number of non-empty blocks as specified by z, n_k is the number of vertices assigned to block k according to z, and $\Gamma(\cdot)$ is the gamma function.

By Bayes's theorem, according to (1.1) and (1.2), the posterior distribution of z given X is

$$p(z \mid X) \propto p(X \mid z)p(z) = \left[\prod_{1 \leqslant k \leqslant l \leqslant K} \frac{B(m_{k,l} + a, \bar{m}_{k,l} + b)}{B(a, b)} \right] \left[\frac{A^K \Gamma(A)}{\Gamma(A + N)} \prod_{k=1}^{K} \Gamma(n_k) \right].$$

We now move to carry out the single site Gibbs method. For every $i \in \{1, 2, \ldots, N\}$, let $z^{(-i)}$ denote the vector $(z_1, z_2, \ldots, z_{i-1}, z_{i+1}, \ldots, z_N)$ obtained from z by removing the i-th entry. We need to compute, for every $i \in \{1, 2, \ldots, N\}$, what's proportional to

$$p(z_i \mid z^{(-i)}, X).$$

Therefore, for each $k \in \{1, 2, \ldots, K^{(-i)}\}$, where $K^{(-i)}$ is the number of non-empty blocks as specified by $z^{(-i)}$, we must calculate $p(z_i = k \mid z^{(-i)}, X)$.

References

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